

Customer Shopping Behavior Analysis

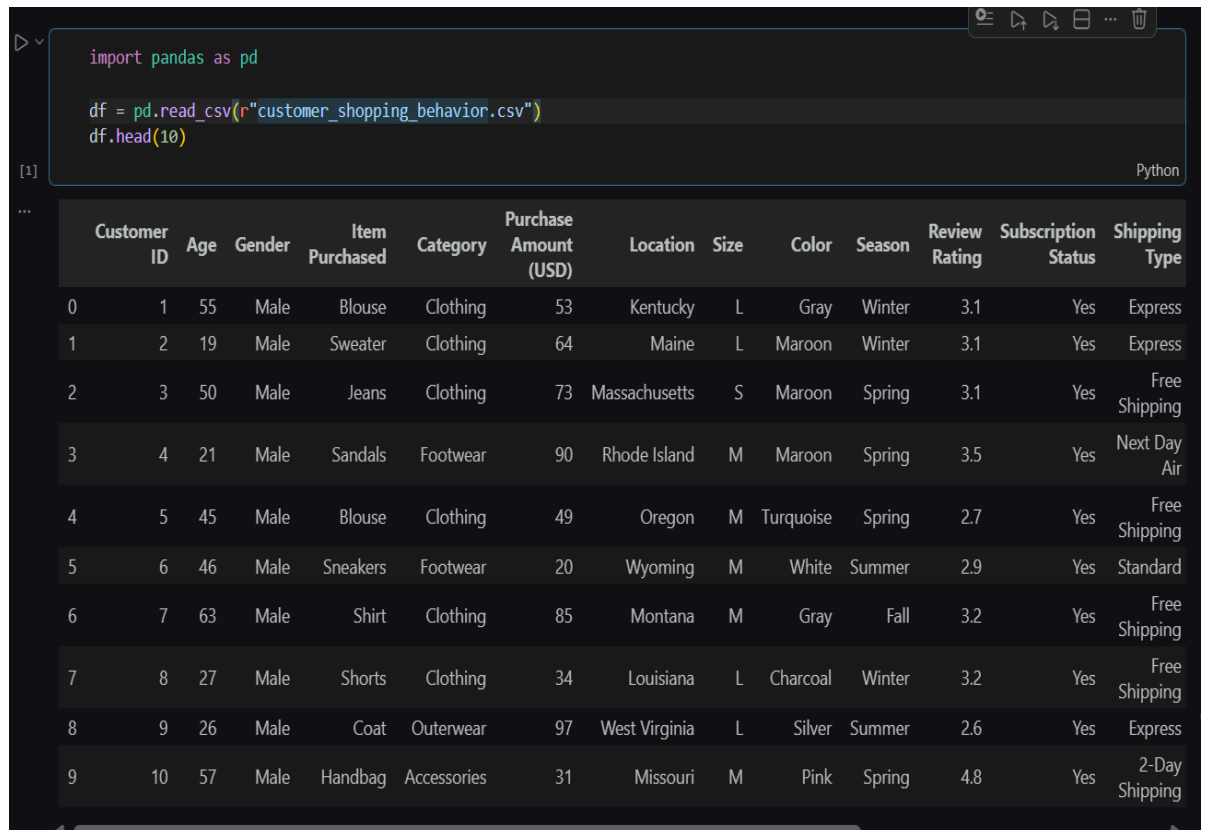
1. Project Overview This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18
- **Key Features:**
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- **Missing Data:** 37 values in the Review Rating column.

3. Exploratory Data Analysis & Modeling (Python) The dataset was prepared, cleaned, and modeled using Python before database integration:

- **Data Loading & Inspection:** Loaded via Pandas, utilizing `info()` and `describe()` to verify dimensions and check data distributions.



```
import pandas as pd

df = pd.read_csv(r"customer_shopping_behavior.csv")
df.head(10)
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping
5	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard
6	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping
7	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping
8	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express
9	10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Customer ID           3900 non-null   int64  
 1   Age                   3900 non-null   int64  
 2   Gender                3900 non-null   object  
 3   Item Purchased        3900 non-null   object  
 4   Category              3900 non-null   object  
 5   Purchase Amount (USD) 3900 non-null   int64  
 6   Location              3900 non-null   object  
 7   Size                  3900 non-null   object  
 8   Color                 3900 non-null   object  
 9   Season                3900 non-null   object  
10  Review Rating         3863 non-null   float64 
11  Subscription Status    3900 non-null   object  
12  Shipping Type          3900 non-null   object  
13  Discount Applied       3900 non-null   object  
14  Promo Code Used        3900 non-null   object  
15  Previous Purchases     3900 non-null   int64  
16  Payment Method         3900 non-null   object  
17  Frequency of Purchases 3900 non-null   object  
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

Missing Data Handling: Imputed missing values in the `review_rating` column using the median rating of each respective product category.

Column Standardization: Cleaned headers into standard snake_case formatting.

Feature Engineering: Created the `age_group` binning column and transformed purchase frequencies into numeric days.

Database Integration: Connected the workflow directly to a relational database to host the clean table for structured querying.

4. Data Analysis using SQL (Business Transactions)

Structured queries were executed against the database to solve key business objectives, split between core operational metrics and advanced custom analytics:

- **Custom Analytics Insights .**
 - **Top Revenue-Generating Locations:** Evaluated geographic performance to isolate top markets and trace average customer ages.

location	total_revenue	avg_customer_age
Montana	5784	43.8
Illinois	5617	43.3
California	5605	42.7
Idaho	5587	43.3
Nevada	5514	43.6

Payment Method Preferences: Tracked transaction volumes and total revenues across gateways like PayPal and credit cards.

payment_method	total_transactions	total_revenue
Credit Card	671	40310
PayPal	677	40109
Cash	670	40002
Debit Card	636	38742
Venmo	634	37374
Bank Transfer	612	36544

Seasonal Category Performance: Cross-referenced seasons and categories to optimize seasonal inventory flows.

season	category	total_revenue
Fall	Clothing	26220
Fall	Accessories	19874
Fall	Footwear	8665
Fall	Outerwear	5259
Spring	Clothing	27692
Spring	Accessories	17007
Spring	Footwear	9555
Spring	Outerwear	4425
Summer	Clothing	23078
Summer	Accessories	19028
Summer	Footwear	9393
Summer	Outerwear	4278
Winter	Clothing	27274
Winter	Accessories	18291
Winter	Footwear	8480
Winter	Outerwear	4562

Discount Impact on Review Ratings: Measured whether discounted items experience variances in customer satisfaction scores.

discount_applied	avg_review_rating	total_purchases
Yes	3.74	1677
No	3.76	2223

Revenue by Gender: Compared total spend metrics between male and female cohorts.

gender	revenue
Male	157890
Female	75191

Subscribers vs. Non-Subscribers: Analyzed average basket sizes and overall revenue generation for active subscribers.

subscription_status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

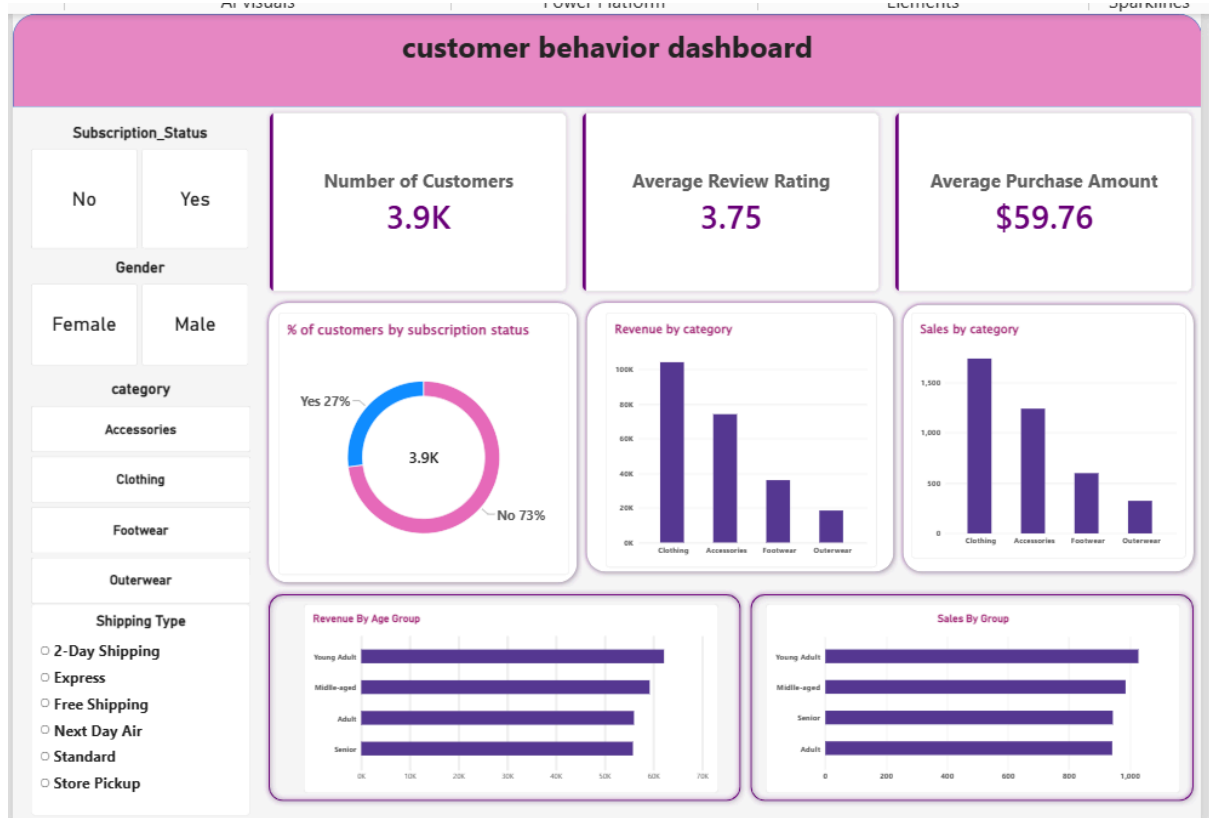
Customer Segmentation: Grouped buyers into *New*, *Returning*, and *Loyal* tiers based on historical purchase counts.

customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

Top Products per Category: Extracted top-ranked items using window functions (`ROW_NUMBER()`) partitioned by category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

5. Visualization & Interactive Dashboard (Power BI) An interactive Power BI dashboard was built to translate the underlying database metrics into executive-ready visual components.



- **Core KPIs:** High-level metric cards track **Total Customers (3.9K)**, **Average Purchase Amount (\$59.76)**, and **Average Review Rating (3.75)**.
- **Behavioral Breakdowns:** Donut charts illustrate subscription distribution (27% Yes vs. 73% No), while column/bar charts outline category sales, revenue performance by age brackets, and geographic concentrations.

6. Business Recommendations

- **Optimize Geographic & Regional Campaigns:** Focus marketing capital on top-performing locations identified in location analytics.
- **Boost Subscription Adoption:** Introduce tiered rewards for non-subscribers to shift them toward high-retention subscription models.
- **Refine Discount Strategies:** Evaluate the margin trade-offs of heavily discounted product lines to ensure long-term profitability.
- **Leverage Lifecycle Segmentation:** Deploy targeted retention workflows customized specifically for *Returning* and *Loyal* customer tiers.